5.4.2 The Cholesky Factorization Theorem

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fit width

We will prove the following theorem in Subsection 5.4.4.

> **Theorem 5.4.2.1. Cholesky Factorization Theorem.** *Given an HPD matrix A there exists a lower triangular matrix L such that $A = LL^H$. If the diagonal elements of L are restricted to be positive, L is unique.*

Obviously, there similarly exists an upper triangular matrix U such that $A = U^H U$ since we can choose $U^H = L$.

The lower triangular matrix L is known as the Cholesky factor and LL^H is known as the Cholesky factorization of A. It is unique if the diagonal elements of L are restricted to be positive. Typically, only the lower (or upper) triangular part of A is stored, and it is that part that is then overwritten with the result. In our discussions, we will assume that the lower triangular part of A is stored and overwritten.